CS 70 — Note 5 Practice Problems

Problem 1: De Bruijn Sequence

A de Bruijn sequence is a 2^n-bit circular sequence in which every binary string of length n occurs exactly once as a contiguous substring. The sequence is generated by an Eulerian tour of the de Bruijn graph.

The de Bruijn graph for $n = 3$. The vertices are the 2-bit strings $\{00, 01, 10, 11\}$. Each vertex a_1a_2 has two outgoing edges: shift left and append 0, or shift left and append 1. The eight edges are:

$$\begin{array}{ll} 00 \to 00 \text{ (append 0)} & 00 \to 01 \text{ (append 1)} \\ 01 \to 10 \text{ (append 0)} & 01 \to 11 \text{ (append 1)} \\ 10 \to 00 \text{ (append 0)} & 10 \to 01 \text{ (append 1)} \\ 11 \to 10 \text{ (append 0)} & 11 \to 11 \text{ (append 1)} \end{array}$$

Since every vertex has in-degree equal to out-degree (both 2), an Eulerian tour exists.

Eulerian tour (starting at 00):

$$00 \to 00 \to 01 \to 10 \to 01 \to 11 \to 11 \to 10 \to 00$$

This uses all eight edges exactly once and returns to the start.

Generating the sequence. Reading off the bit appended at each edge traversal:

$$0,\ 1,\ 0,\ 1,\ 1,\ 1,\ 0,\ 0$$

gives the de Bruijn sequence **00010111**. Reading its eight circular windows of length 3 recovers all eight binary strings $000, 001, 010, 101, 011, 111, 110, 100$, each exactly once.

Problem 2: Completing the Induction of Theorem 5.5

We complete the inductive step of the forward direction of Theorem 5.5, in the case where removing a vertex v from a connected acyclic graph G (on $n = k + 1$ vertices) disconnects it.

Part (a): Two components

Assume G is connected and contains no cycles. Remove an arbitrary vertex $v \in V$ from G along with its incident edges to obtain G'. Suppose G' has two distinct connected components G'_1 and G'_2.

Since both G'_1 and G'_2 are connected and acyclic (removing a vertex cannot create a cycle), we apply the (strong) inductive hypothesis to each. Say G'_1 has s vertices and G'_2 has t vertices; then G'_1 has $s-1$ edges and G'_2 has $t-1$ edges.

We note $s+t=k$, the total number of vertices in G'.

Now add v back to recover G. Since G is connected, v must connect to both components. Moreover, v connects to each component *exactly once*: if v were incident to two vertices of the same component, then since that component is already connected there would already be a path between those two vertices, and adding v would create a second path — hence a cycle, contradicting that G is acyclic. Therefore v contributes exactly 2 edges. The total edge count is:

$$(s-1)+(t-1)+2 = s+t = k = (k+1)-1 = n-1$$

as desired. □

Part (b): $t \geq 2$ components

More generally, suppose removing v splits G' into $t \geq 2$ distinct connected components $G'_1, \ldots, G'_t$, where G'_i has n_i vertices. By the strong inductive hypothesis applied to each component, G'_i has $n_i - 1$ edges, and:

$$n_1 + n_2 + \cdots + n_t = k$$

The total number of edges in G' is therefore $(n_1-1)+\cdots+(n_t-1)$.

Adding v back: since G is connected, v must connect to each of the t components, and by the acyclic argument above it connects to each exactly once. So v contributes exactly t edges. The total edge count of G is:

$$\big[(n_1-1)+\cdots+(n_t-1)\big]+t = (n_1+\cdots+n_t)-t+t = k = (k+1)-1 = n-1$$

as desired. This completes the inductive step of Theorem 5.5. □

Problem 3: Connected Graphs Have at Least $n-1$ Edges

Claim: Any connected graph on n vertices has at least $n-1$ edges.

Proof by strong induction on $n = |V|$.

Base case. $n=1$: a single vertex has 0 edges, and $0 \geq 1-1 = 0$. ✓

Inductive hypothesis. Assume every connected graph with m vertices, for all $1 \leq m \leq n$, has at least $m-1$ edges.

Inductive step. Let G be a connected graph on $n+1$ vertices. Remove an arbitrary vertex v along with its incident edges, yielding a graph G' that splits into $t \geq 1$ connected components $G'_1, \ldots, G'_t$, where G'_i has n_i vertices with $n_i \leq n$.

By the inductive hypothesis applied to each component, G'_i has at least $n_i - 1$ edges. The vertices satisfy:

$$n_1 + n_2 + \cdots + n_t = n$$

Since G is connected, v must be incident to at least one vertex in each of the t components (otherwise some component would be unreachable from v). Thus v contributes at least t edges. The total number of edges in G is at least:

$$\big[(n_1 - 1) + \cdots + (n_t - 1)\big] + t = (n_1 + \cdots + n_t) - t + t = n = (n+1) - 1$$

Therefore G has at least $(n+1) - 1$ edges, completing the induction. $\square$